Don Fox
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https://resume-site-b43d.onrender.com/resume

Summary:

Thirty years of professional software development spanning telecommunications, healthcare IT, and data systems. Primary focus has been backend engineering on Unix/Linux platforms, with experience across a broad range of languages, databases, and system architectures. More recent work includes web development, functional programming, and AI-assisted development and prompt engineering.

Proof-of-concept development — particularly in healthcare IT — has been a recurring part of the work, as has intern mentoring and occasional collaboration with academic and research teams. Interested in expanding on that going forward.

Education:

- **Bachelor of Arts in Philosophy, 1982**
  University of Tennessee, Knoxville TN

Technical Training

- **Associate of Science Degree in Computer Science Technology, *Cum Laude,* 1991**
  Pellissippi State Technical Community College, Knoxville TN

Technical Skills

Languages: Python, C, C++, Perl, Elixir, Haskell, PL/SQL, C#
Frameworks: Django, Flask, Phoenix, SQLAlchemy
Databases: PostgreSQL, MySQL, Oracle, MongoDB, MariaDB, Redis, Informix, DB2
Platforms: Linux, Unix, macOS, Solaris, AIX, Windows
Tools & Technologies: Git, Docker, Node.js, Shell Scripting, RESTful APIs

Professional History:

EMR Technical Solutions (EMRTS) — *Software Engineer (Independent Contractor)*
https://emrts.us/

Independent contractor working with a healthcare IT startup across six engagements, developing open-source tools for medical records, billing, provider enrollment, and claims. Work has centered on proof-of-concept development and R&D, with occasional collaboration with academic and professional partners. Contracts listed below fill the gaps between EMRTS engagements.

- July 2025–August 2026: Developed a proof-of-concept Provider Vault medical enrollment website, combining Elixir/Phoenix for scalability and concurrency with Python for AI-assisted development and prompt engineering. Subsequently developed the Medicaid Claims Checker, which pairs a Haskell rule engine with an Elixir/Phoenix front end (web UI, X12/837P claim parsing, provider validation, and batch processing), with an integration point for external ML risk scoring; claims are flagged for human review before a determination is finalized. Submitted via RFI (Request for Information) to several state governments. Mentoring interns on complementary healthcare IT projects, with topics including machine learning and prompt engineering. Tech: Haskell, Elixir, Phoenix, Python 3.11, PostgreSQL
- August 2024–December 2024: Collaborated with interns on a Medicaid prediction research project, introducing foundational data science concepts and workflows. Led a follow-up initiative exploring cryptocurrency, developing a Bitcoin indexer prototype. Tech: Python 3.x, Flask, PostgreSQL
    - *Interim contracts: CSpire, Jackson MS (July 2018–January 2019); Orchid Orthopedic Solutions, Remote (February 2020–December 2020); Quantuvos, Remote (July 2021–February 2022) — see entries below.*
- July 2015 – January 2017: Developed proof-of-concept tools and prototypes for processing medical transaction data (EDI_X12_837) and handling large medical provider datasets (JSON, XML). Extended a web services application for state governments and clinics to facilitate medical provider data registration and validation.
  Tech: Python 3.5, PostgreSQL, Django, JavaScript | Platform: Unix
- October 2013 – March 2015: Developed research-driven projects for mobile medical screenings and provider enrollment for state governments. Contributed to developing a Health Information Exchange (HIE) application for hospitals. Tech: Elixir/Phoenix | Platform: Ubuntu Linux, macOS | Databases: MariaDB, PostgreSQL, MongoDB
    - Interim contracts: Hilton, Memphis TN (April 2012–August 2012); Lowe's, Mooresville NC (August 2012–June 2013) — see entries below.
- January 2012 – March 2012: Focused on electronic medical records, claims processing, and provider enrollment applications on GNU/Linux. Developed tools for handling JSON data and extended the application's support for additional records processing.
  Tech: Python, Django, JavaScript (jQuery, Node.js) | Databases: MariaDB, PostgreSQL, Redis
    - *Interim contract: Hewlett-Packard, Nashville TN (May 2010–December 2011) — see entry below.*

- July 2009 – May 2010: Conducted code analysis for GNUmed, an open-source electronic medical record system, to evaluate its suitability for client needs. Built a RESTful web service for state medical provider validation.
  Tech: Python, Django | Platform: GNU/Linux

Quantuvos — *Technical Consultant (Contract)*
Remote | July 2021 – February 2022
Technical consulting for Quantuvos, a coaching-platform startup. Reviewed and evaluated code for the client-facing MVP, supporting coaches and clients in scheduling workshops and sessions, identifying issues and recommending improvements. Contract position. **Tech:** Python 3.9, Django 3.1, PostgreSQL 14.1, Docker

Orchid Orthopedic Solutions — *Python Developer (Contract)*
Remote | February 2020 – December 2020 https://www.orchid-ortho.com
Contract work preparing Word, Excel, and PDF documents for archival submission to EtQ Reliance. Programmatically inserted tags and formatting into them and validated spreadsheet metadata for accuracy and completeness. Contract position. **Tech:** Python 3.8, C#, Windows/macOS

CSpire — *Python Developer (Contract)*
J**ackson, MS | July 2018 – January 2019** https://www.cspire.com
Led development of a RESTful auto-provisioning API for tracking and managing customer services delivered through CSpire and its third-party vendor partners. The application supported logging, reporting, and administration for both business office and network teams. Contract position. **Tech:** Python 2.7, Flask, SQLAlchemy, MySQL, PostgreSQL

Lowe's — *C Developer (Contract)*
Mooresville, NC | August 2012 – June 2013 https://corporate.lowes.com
Refactored and extended legacy C code to support EMV (chip-based credit and debit card) transaction processing as part of Lowe's transition away from magnetic stripe (swipe) cards. Work included code analysis, bug fixes, and documentation. Contract position.

Hilton — *Production Support Developer (Contract)*
Memphis, TN | April 2012 – August 2012
Provided production support for Hilton's Central Reservation System, contributing bug fixes and maintenance using C, Perl, and shell scripting. Contract position. **Tech:** Solaris, GNU/Linux, Informix, MySQL

Hewlett-Packard — *Software Engineer*
Nashville, TN | May 2010 – December 2011
Performed software maintenance and troubleshooting for TennCare (State of Tennessee). Key projects included:
- Refactored and extended a SOAP server (written in C) for enhanced data exchange within the Medicaid Management Information System (MMIS).
- Developed monthly report generators for medical claims processing.
- Ported the entire application codebase to a new 64-bit Solaris network with updated compilers and Oracle database version.
- Contributed to technical writing and documentation.

Tech: C, PL/SQL, SOAP, Oracle, Solaris

ACS — *Software Developer (Contract)*
Nashville, TN | February 2009 – June 2009
Migrated MMIS (Medicaid Management Information System) software for TennCare from Hewlett-Packard's infrastructure to ACS (Affiliated Computer Services). Developed automation scripts and analyzed legacy C modules. Contract position.

Nationwide — *Python Developer / Data Analyst (Contract)*
Columbus, OH | October 2008 – February 2009 Analyzed annuity and financial data daily to detect anomalies in high-volume datasets. Proposed and developed a Python-based data validation and reporting tool to modernize and replace aging shell scripts used in production. Contract position. **Tech:** Solaris, Oracle, proprietary DSL

Lexmark — *C Developer (Contract)*
Lexington, KY | May 2007 – October 2008 https://www.lexmark.com/en_us.html
Bug fixes, code analysis, and refactoring of C code for high-end prototype printers designed for secure environments (government, military, finance). Developed Python scripts for daily integration testing. Contract position. **Tech:** C, Python 2.6, RHEL

IBM Internet Security Systems — *Build Tools Developer (Contract)*
Atlanta, GA | January 2007 – May 2007
Developed tools for porting and rebuilding C/C++ applications from Microsoft Visual Studio to Unix environments, using Visual Studio build artifacts rather than traditional makefile-based builds. Contract position. **Tech:** C/C++, Visual Studio, macOS, Ubuntu Linux, Windows XP

Lowe's — *C Developer (Contract)*
Wilkesboro, NC | February 2006 – December 2006
Adapted Lowe's point-of-sale software for the company's first Canadian store, modifying C code for local currency, card processing, and regulatory compliance. Contract position.
Tech: C, AIX, Oracle, DB2

Cingular Wireless — *C++ Developer (Contract)*
Alpharetta, GA | August 2004 – February 2006
Maintained and extended a legacy billing engine running on AIX and GNU/Linux using Oracle. The system was written primarily in C++ with shell scripts. Taught a C++ course to the code acceptance testing group and developed a Perl application that automated the configuration of their test boxes. Contract position. **Tech:** C++, Perl, Shell, Oracle, AIX, GNU/Linux

SAIC (Science Applications International Corporation) — *Perl Developer (Contract)*
Oak Ridge, TN | February 2003 – August 2004 https://saic.com
Maintained and extended a legacy Perl application for network-wide resource and account management at Oak Ridge and Los Alamos National Laboratories until it was decommissioned. Contract position. **Tech:** Perl, Oracle, Unix

University of Tennessee – Center for Environmental Biotechnology — *Bioinformatics Programmer (Independent Contractor)*
Knoxville, TN | August 2002 – December 2002 https://ceb.utk.edu/ Built bioinformatics tools in Perl and C, backed by MySQL on Red Hat Linux, for monitoring, reporting, and analysis of microarray gene data. Tutored professional staff and graduate students in Perl programming for bioinformatics research. Independent contract position. **Tech:** Perl, C, MySQL, Red Hat Linux

Manhattan Associates — *C++ Developer (Contract)*
Atlanta, GA | July 2001 – July 2002
Performed code maintenance on a cross-platform warehouse management system written in C++ for Linux and Windows, using Oracle for data storage and the MKS Toolkit for Unix-style tooling on Windows. Contract position. **Tech:** C++, Oracle, Linux, Windows, MKS Toolkit

LHS/Sema Group Telecom — *C/C++ Developer (Contract)*
Atlanta, GA | January 2000 – July 2001
Performed code maintenance and enhancement for a large legacy billing engine written in C/C++ using Oracle, spanning multiple Unix platforms. Contributed to major technical documentation rewrites. Contract position. **Tech:** C, C++, Oracle, Unix

Intellivoice / Premier Technologies — *R&D Software Developer (Contract)*
Atlanta, GA | November 1998 – January 2000
Worked on multimedia integration R&D for distributed Solaris networks. Developed coding projects using C++ and evaluated third-party middleware (such as CORBA) to facilitate network communication. Contract position. **Tech:** C++, CORBA, Solaris

MCI/WorldCom — *Software Developer (Contract)*
Atlanta, GA | February 1996 – November 1998
Full-cycle project to implement a billing engine on AIX using C++, Pro*C, and Oracle. Developed a Perl application that processed firearm purchase applications for the State of California (an MCI client). Contract position. **Tech:** C++, Pro*C, Oracle, SQL, AIX, Unix, Perl